

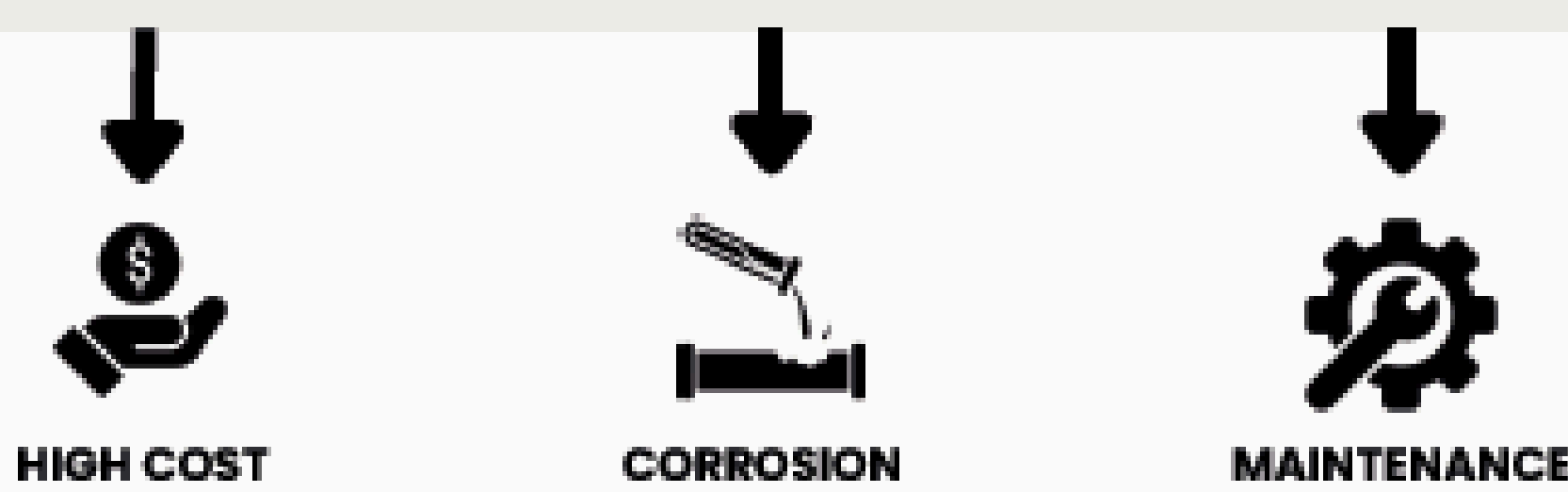
Enhanced Corrosion Resistance of Aluminum Anodes in Aluminum-Air Batteries using Multilayered PVD Coatings

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PROBLEM STATEMENT

The battery functions by oxidizing aluminum, practical implementation remains significantly hindered by the rapid corrosion and passivation of aluminum anodes. These phenomena result in substantial parasitic hydrogen evolution, poor coulombic efficiency, shortened operational lifespan, and increased maintenance costs due to frequent component replacements.



OBJECTIVES

- Design and fabricate multilayered PVD coatings for aluminum anodes in AABs, achieving a better corrosion resistance than uncoated aluminum using standard tests.
- Characterize the microstructure, composition, and electrochemical properties of coated aluminum anodes in AABs to evaluate performance.
- Evaluate the performance of coated aluminum anodes by measuring discharge capacity, energy density, and corrosion rate using standard testing procedures.
- Optimize PVD coating parameters within 8 weeks to achieve an improvement in corrosion resistance and discharge capacity of aluminum anodes in aluminum-air batteries.

BACKGROUND

Aluminum-air batteries are promising for high-energy-density storage due to aluminum's abundance and high theoretical energy density (~8.1 kWh/kg). However, rapid corrosion of aluminum anodes in aqueous electrolytes, especially in alkaline or saline environments, limits their lifespan and efficiency. To address this, protective surface modifications using Physical Vapor Deposition (PVD) techniques, particularly multilayer coatings, are being explored. These coatings enhance corrosion resistance by blocking ionic diffusion and crack propagation, while maintaining electrochemical activity. By optimizing multilayer structures with corrosion-resistant and sacrificial materials, the operational life and performance of aluminum-air batteries can be significantly improved, supporting their commercial use in off-grid, backup, and portable energy applications.



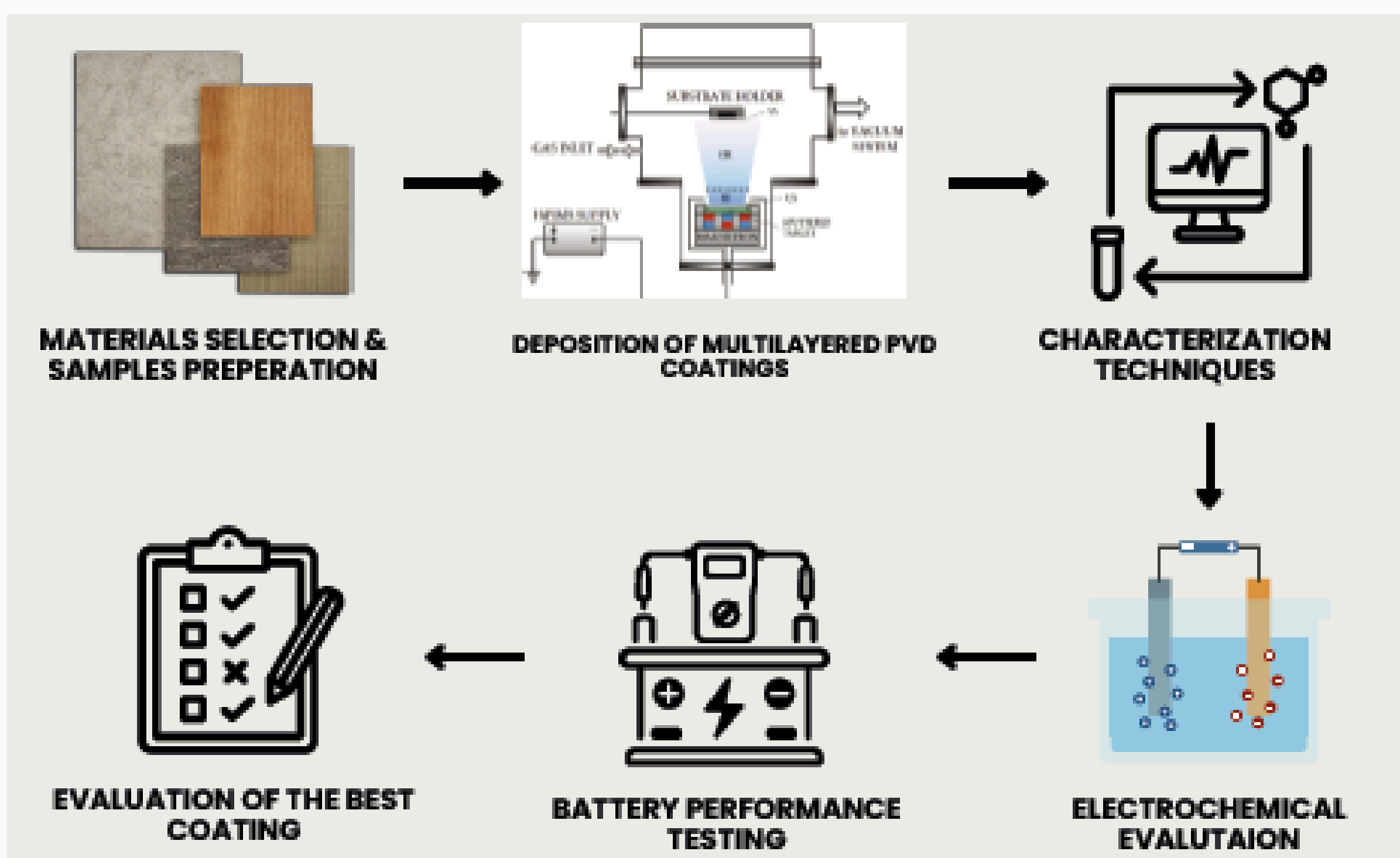
NOVELTY

This research presents a novel approach to enhancing Aluminum-Air Batteries (AABs) by applying multilayered PVD coatings to aluminum anodes. The coatings aim to improve corrosion resistance by at least 30%, extending anode lifespan and reducing maintenance costs. The study also optimizes coating parameters to boost both corrosion resistance and discharge capacity. Through microstructural and electrochemical characterization, this work advances AAB efficiency and supports the development of more sustainable battery technologies.

HYPOTHESIS

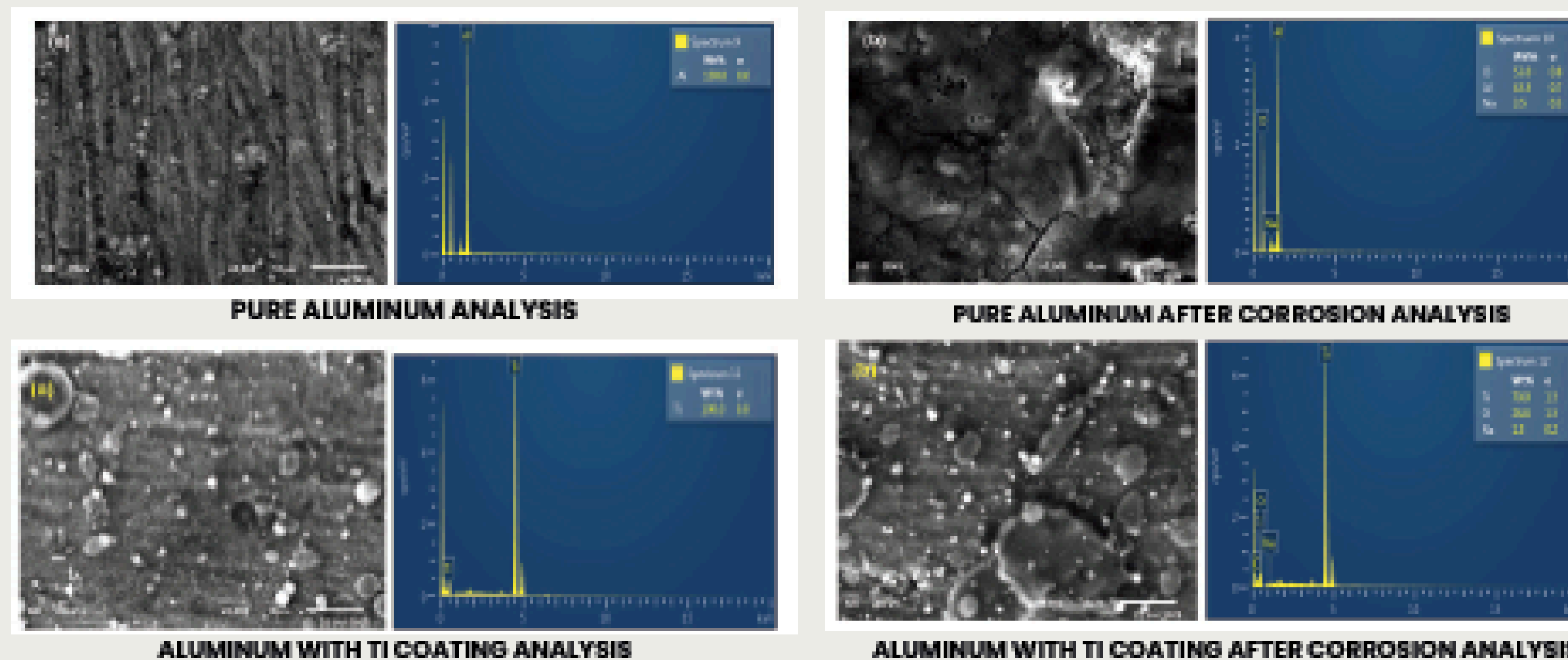
PVD multilayer coatings will enhance the corrosion resistance and electrochemical performance of aluminum anodes, improving the efficiency and lifespan of aluminum-air batteries.

METHODOLOGY

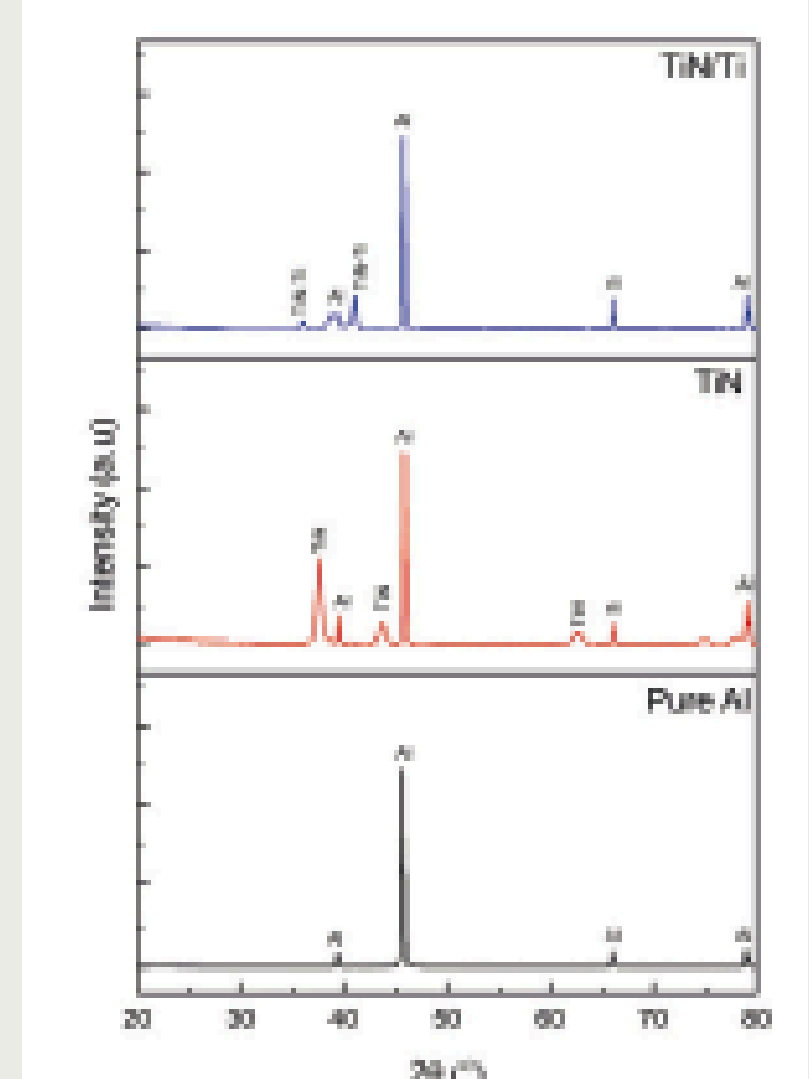


RESULTS

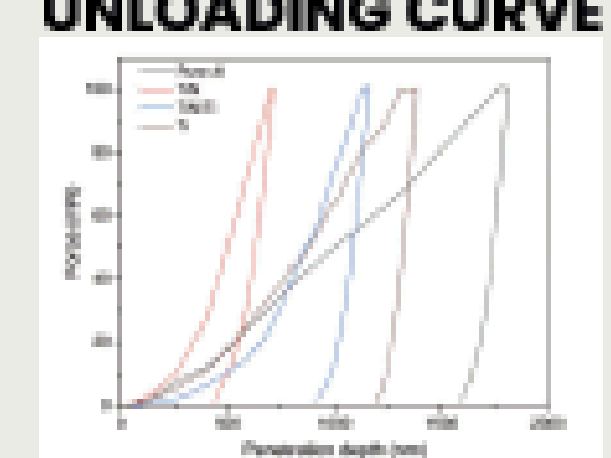
CORROSION TEST USING 4M NaOH



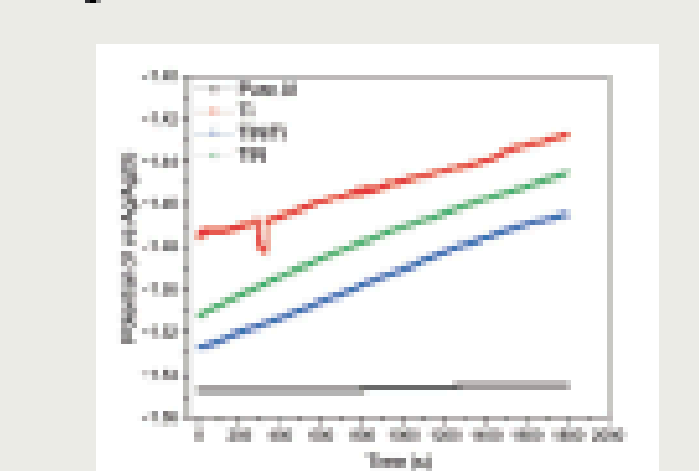
XRD TEST OF COATINGS



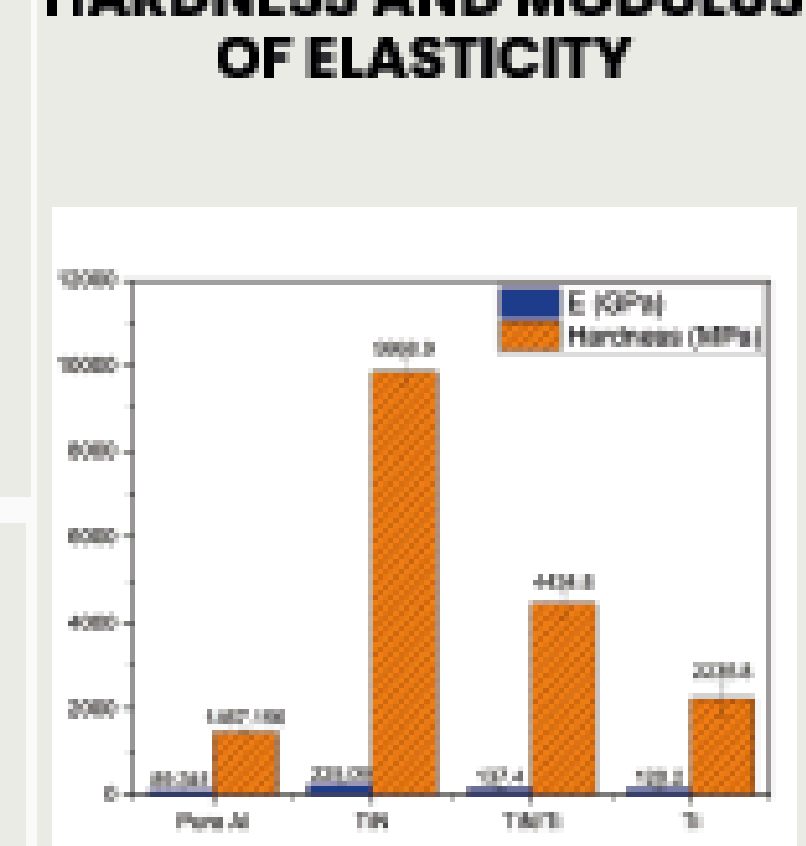
LOADING AND UNLOADING CURVE



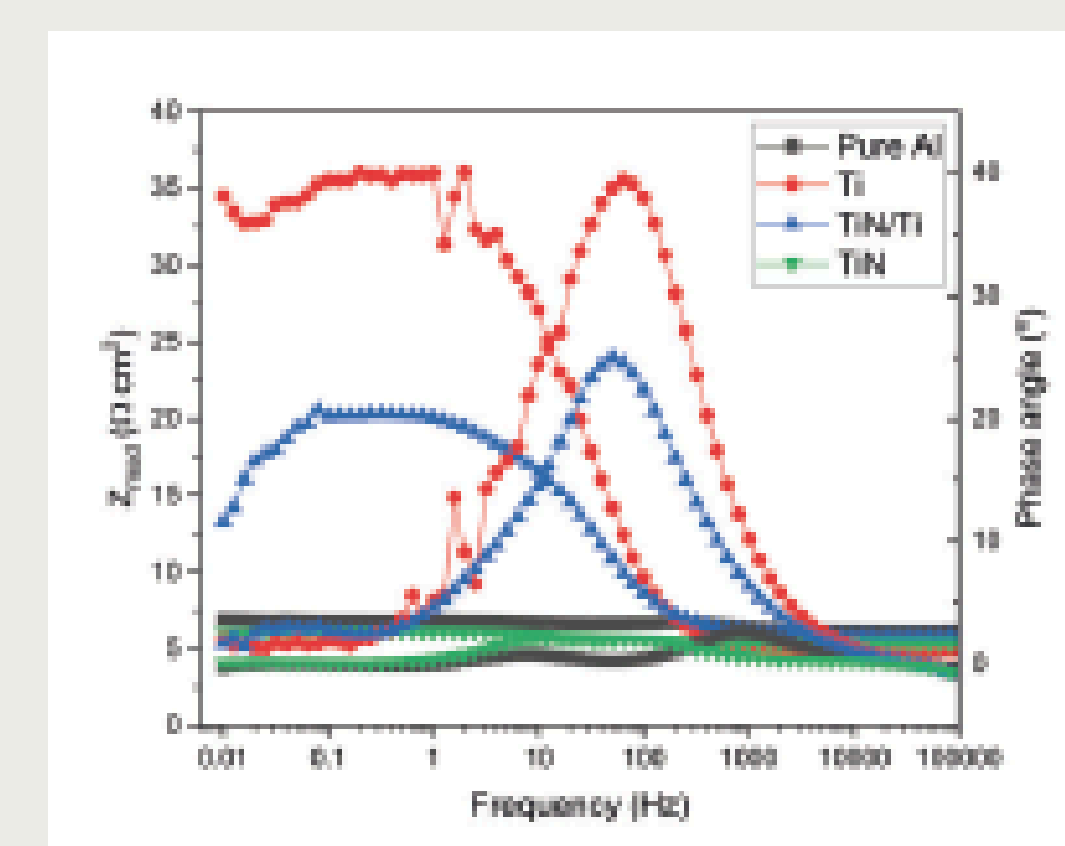
Open Circuit Potential



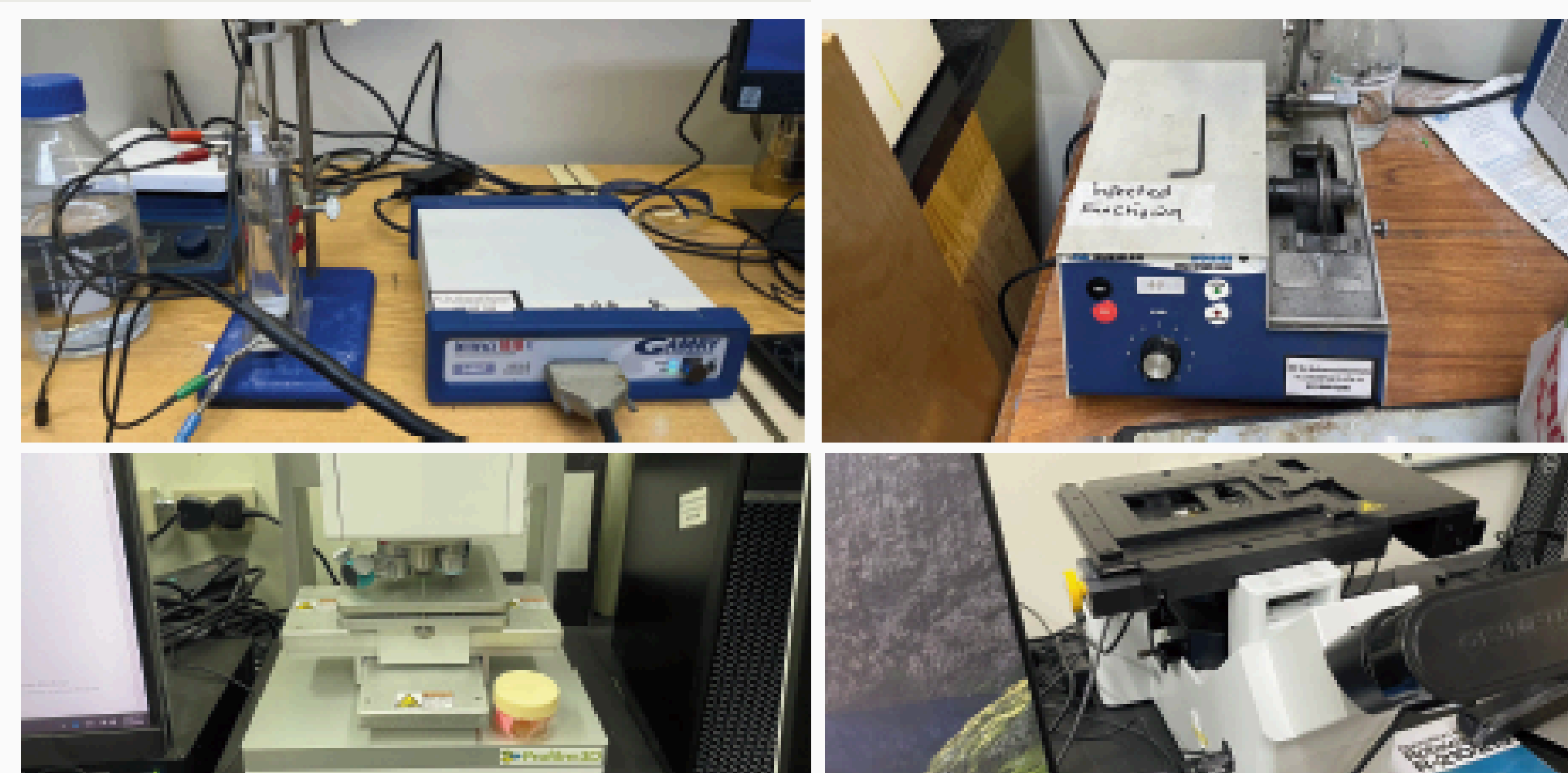
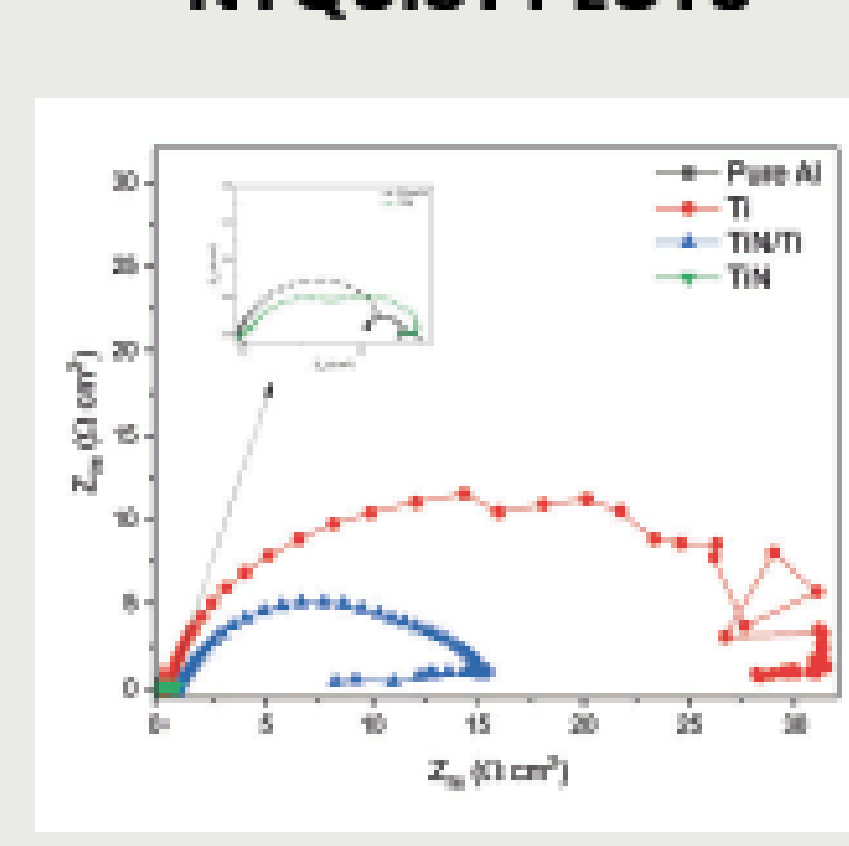
HARDNESS AND MODULUS OF ELASTICITY



Bode and Phase Plot



NYQUIST PLOTS



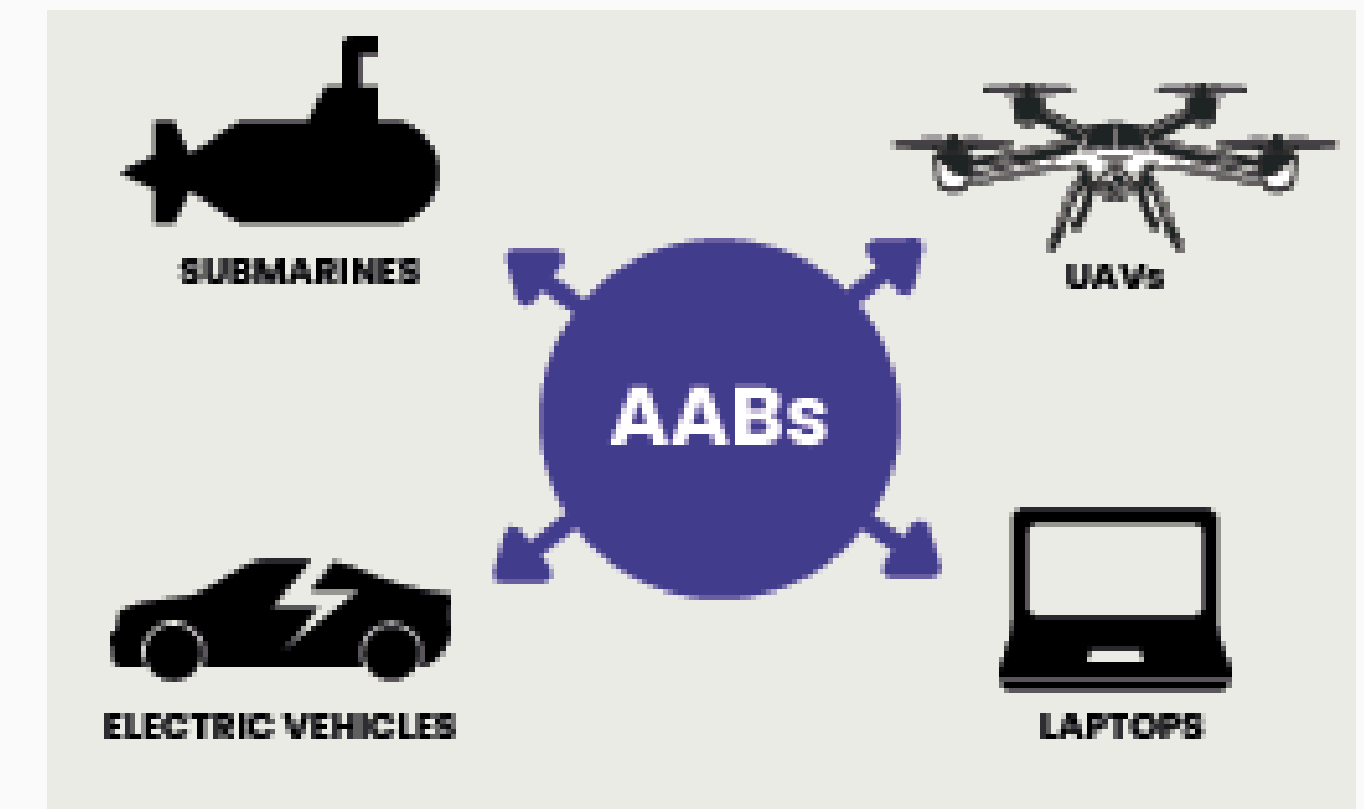
IMPACT

This research improves Aluminum-Air Battery (AAB) performance by enhancing aluminum anode corrosion resistance with PVD coatings, increasing longevity. Optimizing the coatings leads to more efficient and cost-effective energy storage, advancing battery technology and electrochemical engineering.

CONCLUSION

This research demonstrates the potential of multilayered PVD coatings to significantly improve the performance of aluminum anodes in AABs. By enhancing corrosion resistance and discharge capacity, the study contributes to making AABs more durable, efficient, and commercially viable. The optimization of coating parameters provides a pathway to more sustainable and cost-effective energy storage solutions.

APPLICATION



FUTURE WORK

focus on extended battery performance testing under varied conditions to assess long-term stability and efficiency. Efforts will also aim to enhance the PVD coatings by optimizing deposition parameters and exploring advanced materials to further improve corrosion resistance and electrochemical performance.

REFERENCES

- [1] M. Mokhtar, M.Z.M. Talib, E.H. Majlan, S.M. Tasirin, W.M.F.W. Ramli, W.R.W. Daud, J. Sahari, Recent developments in materials for aluminum-air batteries: A review, J. Ind. Eng. Chem. 32 (2015) 1-20. <https://doi.org/10.1016/j.jiec.2015.08.004>.
- [2] P. Wu, S. Wu, D. Sun, Y. Tang, H. Wang, A Review of Al Alloy Anodes for Al-Air Batteries in Neutral and Alkaline Aqueous Electrolytes, Acta Metall. Sin. (English Lett. 34 (2021) 309-320. <https://doi.org/10.1007/s40195-020-01140-x>.
- [3] Y. Chung, S. Lee, S. Yoon, W.K. Choi, S.K. Jeong, Mitigating anodic corrosion in aluminum-air batteries: Effects of alkali metal cation size on electrochemical performance, Appl. Surf. Sci. 681 (2025) 161533. <https://doi.org/10.1016/j.apsusc.2024.161533>.